Activity - 8 (7)

Retail Sales

Given R code

```
set.seed (6795)
m <- 4 # observations per cycle (quaterly data)
n <- m * 10 # 10 full cycles
t <- 1:n
trend <- 107.9 + 1.041 * t
sales <- round( trend + 22.2 * cos( 2 * pi * (t-4)/m) + rnorm
(n, 0, 6.1, 1)
x <- ts(sales, frequency = m, start = c(2013, 1))
```

① Plot the series & identify the trend

R code

```
plot(x, main = "Quaterly Retail Sales", xlab = "years",
ylab = "Sales (lakh rupees)")
fit <- lm (sales ~ t)
coef (fit)
```

n = 40, cycle length = 4 full cycles = 10

Liniear model coefficients (104.8508), t = 1.1572.

Trend slope = 1.1572 lakh rupee per quartes.

② Seasonal pattern

```
d <- decompose (x)
d$figure
```

Peak quarter Q_4 = 20.3306

Trough " Q_2 = -19.6014

Seasonal swing = peak - trough

= 20.3306 - (-19.6014) = 39.9319 lakh rupees

③ library(ggplot2)

```
df <- data.frame(sales = as.numeric(x),
                 cycle = rep(1:10, each=4),
                 Position = rep(1:4, times=10)).

ggplot(df, aes(x= Position, y = Sales, group = cycle,
               colour = factor(cycle))) +
geom_line() + geom_position() +
labs(title = "seasonal Plot of Quarterly Retail Sales",
     x= "Quarter Position", y = "Sales (lakh rupees)",
     colour = "cycle")
```

④ detrended <- resid(lm(sales ~t))
acf(detrended, lag.max = 4, plot = FALSE)
2 / sqrt(n)

⑤
```
first_cycle <- sales[1:4]
last_cycle <- sales(37:40)
range_first <- max(first_cycle) - min(first_cycle)
range_last <- max(last_cycle) - min(last_cycle)
range_first
range last
range_last / range_first
```

First cycle range = 30.3 lakh rupees
last cycle rang = 53.6 lakh rupees
Ratio (last/first) = 1.7690

DSA06 DATA HANDLING AND VISUALIZATION | Activity: Identify Seasonality Patterns | Slot C

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Time: 20 to 25 minutes Total: 20 marks Tool: R (ggplot2 / base R). Work alone; every student's series is different. Write your answers and paste your plots and code.

Series SP-C15: Retail sales [20 marks]

Scenario: You are the analyst for a retail chain. The series sales (Sales (lakh rupees)) is recorded quarterly. Management believes it follows a repeating pattern and wants it identified before planning. The cycle length is 4 observations. Run the code below to create the series x and answer the tasks.

Evidence: run this code in R (do not edit it).

```
set.seed(6795)
m <- 4                      # observations per cycle (quarterly data)
n <- m * 10                 # 10 full cycles
t <- 1:n
trend <- 107.9 + 1.041 * t
sales <- round(trend + 22.2 * cos(2 * pi * (t - 4) / m) + rnorm(n, 0, 6.1), 1)
x <- ts(sales, frequency = m, start = c(2013, 1))
```

Tasks:

1. Look at the series. Plot x (plot(x) or ggplot2). State n, the cycle length, the number of full cycles, and whether the overall trend is rising, falling or flat (fit lm(sales ~ 1:n) and report the slope per quarter). [3]
2. The seasonal pattern. Use decompose(x) and extract $figure (the 4 seasonal values). Name the peak quarter and the trough quarter with their values, and the seasonal swing (peak minus trough). [4]
3. Is it consistent? Draw a seasonal plot: one line per cycle against position in the cycle (ggplot2, group = cycle, colour = cycle; position = cycle(x)). In how many of the 10 cycles is the highest value in the peak quarter from Task 2? Is the pattern consistent? [3]
4. Confirm with autocorrelation. Detrend (d <- resid(lm(sales ~ 1:n))) and compute acf(d, lag.max = 4, plot = FALSE) using observation lags. Report the autocorrelation at lag 4 and at lag 2, and compare with the significance bound 2/sqrt(n). What do the two values say about seasonality? [4]
5. Additive or multiplicative? Compute the range (max minus min) of sales in the first cycle and in the last cycle and their ratio (last / first); decide the type and justify [2]. Remove the seasonal part with decompose(x, type = ...) using your chosen type (subtract for additive, divide for multiplicative), fit lm(adjusted ~ 1:n) and report the slope per quarter [2]. Write one sentence of advice to management based on your findings [2]. [6]

***Remember:** seasonality repeats at a fixed, known period; a trend is not a season; if the swings grow with the level, the season is multiplicative.*

Code of Conduct: I certify that this work is my own. Signature: _____